

QBitOpt: Fast and Accurate Bitwidth Reallocation during Training

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Abstract

Quantizing neural networks is one of the most effective methods for achieving efficient inference on mobile and embedded devices. In particular, mixed precision quantized (MPQ) networks, whose layers can be quantized to different bitwidths, achieve better task performance for the same resource constraint compared to networks with homogeneous bitwidths. However, finding the optimal bitwidth allocation is a challenging problem as the search space grows exponentially with the number of layers in the network. In this paper, we propose *QBitOpt*, a novel algorithm for updating bitwidths during quantization-aware training (QAT). We formulate the bitwidth allocation problem as a constraint optimization problem. By combining fast-to-compute sensitivities with efficient solvers during QAT, *QBitOpt* can produce mixed-precision networks with high task performance guaranteed to satisfy strict resource constraints. This contrasts with existing mixed-precision methods that learn bitwidths using gradients and cannot provide such guarantees. We evaluate *QBitOpt* on ImageNet and confirm that we outperform existing fixed and mixed-precision methods under average bitwidth constraints, commonly found in literature.

1. Introduction

In the last decade, neural networks have driven some of the most significant advances in artificial intelligence. However, searching for better-performing neural networks has increased their resource requirements substantially. The increase in parameters and training time is most notable in transformer models that are becoming ubiquitous in computer vision [12, 39] and natural language understanding [34, 5]. As a result, deploying these ever-growing neural networks on resource-constrained devices, such as mobile phones, embedded systems, and IoT devices, remains a challenge. Quantization has been proven to be a very ef-

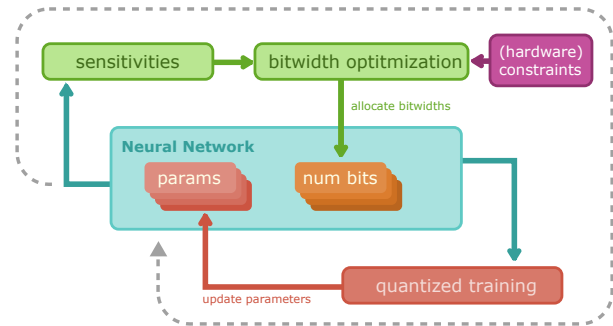


Figure 1: QBitOpt uses a quantization sensitivity metric to infer bitwidths while taking (hardware) constraints into account. The bitwidth allocation is used within a QAT training loop.

fective method of addressing this constraint by reducing the memory and computational cost of neural networks without sacrificing their accuracy [22, 26, 29]. This is achieved by compressing the floating-point weights and activation to a more efficient low bitwidth fixed-point representation. Deploying quantizing neural networks on devices is also becoming easier as increasing number of devices support efficient low-precision operation.

Neural network quantization is an active field of research, and various methods have emerged in recent years. Broadly, we can distinguish them into two main groups: *post-training quantization* (PTQ) [31, 43, 28, 15, 27, 16], and *quantization-aware training* (QAT) [24, 14, 25, 30, 40, 23]. In PTQ, we take a pre-trained neural network as input and infer the quantization setting that maximizes task performance without access to labeled data or the original training pipeline. In contrast, in QAT, we simulate quantization during training and allow the network parameters to adjust to quantization noise. In both cases, the target quantization bitwidth is commonly determined beforehand and is set homogeneously over the entire network. A crucial drawback of this approach is that the homogeneous bitwidth can only be as low as the network’s most sensitive layer allows.

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